

Notice of Intent to Sole Source: MESO QuickPlex SQ 120MM Electrochemiluminescence
Multiplexing Instrument

(75N98026Q00176)

The National Institutes of Health intends to award a Firm-Fixed-Price purchase order to Meso Scale Diagnostics LLC.

In accordance with FAR Part 6.302-1, Meso Scale Diagnostics LLC is the only known responsible source who can provide the item that meets our standards. No other source has been identified who can provide the requirements listed below.

The National Institutes of Health (NIH), National Institutes of Diabetes and Digestive and Kidney Disease (NIDDK), Phoenix Epidemiology and Clinical Research Branch (PECRB) need to obtain a Electrochemiluminescence multiplexing instrument. This instrument is needed to provide multi-biomarker analyses on numerous clinical and research biospecimens in a concise, accurate and rapid fashion. These analyses will be in support of PECRB research protocols.

The following product features/characteristics are required for this requirement:

- Multiplex up to 10 assays/well
- Detection limits of 0.5 – 1.0pg/mL
- Dynamic range greater than 10^5
- Operating temperatures of 20°C – 26°C
- Plate read time of 2 min 45 sec
- Read 22 plates/hour
- Flexibility of processing single and multiplex assays
- Compatible with SECTOR™ and QuickPlex products
- Compact size
- No fluidics
- No daily, monthly user maintenance
- Touchscreen laptop computer
- Limited one-year warranty

Only the requested system will meet the government's needs because it provides a unique combination of sensitivity, multiplexing capability, and low sample volume requirements that are essential for the ETCHED longitudinal observational study.

The ETCHED study requires simultaneous measurement of a broad panel of autoimmune and metabolic biomarkers across maternal, cord blood, and infant samples. These samples—particularly from infants—are extremely limited in volume, with approximately 5 milliliters needing to support both clinical care and research testing. Additionally, the study includes samples from indigenous pediatric populations, where shipping specimens off-site presents logistical, safety, and integrity risks. The compact design and on-site analytical capability of the QuickPlex 120 eliminate the need for off-site processing, thereby preserving sample integrity, reducing risk of loss or degradation, and ensuring timely analysis. Other available technologies would either require larger sample volumes, lack sufficient multiplexing capability, or necessitate off-site processing, all of which would compromise study feasibility and data quality.

Without this system, the institute cannot perform the comprehensive autoimmune and metabolic marker analyses required for pediatric and infant research participants. Existing methods are inadequate to process the necessary number of analytes within the strict sample volume

constraints. Failure to obtain this capability would significantly impair the institute's ability to generate critical data supporting pediatric diabetes and metabolic disease research.

The SECTOR S 600 and QuickPlex® SQ 120 laboratory analytical instruments are manufactured and sold solely by Meso Scale Discovery (MSD), a division of Meso Scale Diagnostics, LLC of Gaithersburg, MD. The Imagers are compatible only with microwell plates and immunoassay kits sold by MSD® and vice versa.

MSD utilizes electrochemiluminescence to detect binding events on patterned arrays. There are no other technologies that enable multiplexed measurement using electrochemiluminescence for detection. And there are no other suppliers that provide an electrochemiluminescence detection system that is not reliant on beads or flow cells. Therefore, the QuickPlex 120 represents the only solution capable of meeting the government's minimum requirements for sensitivity, efficiency, sample preservation, and operational feasibility within this study.

This notice is not a request for competitive quotes. Meso Scale Diagnostics LLC is the manufacturer. However, any available small businesses (e.g., 8(a), service-disabled veteran owned small business, HUBZone small business, small disadvantaged business, veteran-owned small business, and women-owned small business) as stated herein may submit a written capability statement that clearly supports and demonstrates their ability to perform the requirement. A determination by the Government to compete this proposed award based upon responses to this notice is solely within the discretion of the Government.

It is anticipated that an award will be issued approximately five (5) days after the date of this notice unless the Government determines that another organization has the capability to meet this requirement.

Please email responses to david.shen@nih.gov no later than 7:00 a.m. EST on April 15, 2026.